Maritime "Shadow Fleet" Detection with Parallel Computing

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Agenda

- Introduction
- Parser
- Results
- Challenges
- Profiling
- Real world analysis

Introduction

- Investigate ship anomalies in 2025-02-28 and 2025-03-01 date **Danish Maritime Authority** AIS datasets applying parallel computing
- To execute the script across multiple platforms in order to evaluate its capabilities under different sets of configurations
- Perform testing on platforms, such as Intel i7 8 core processor, MAC M2 Chip 8 cores

File parser

- Dataset .csv file reading is performed by chunks and in parallel way using Python multiprocessing pool
- Multiprocessing Pool automatically creates workers and distributes the workload to them
- The data is read and written in an unordered way to a local *SQLite* database file, which used for ship data anomaly detection
- When reading is required, we sort the data in a SQL query and our Python script fetches the data by chunks using workers during parallel reading
- The parser works independently of other parts of the code. Parallelism was applied to each parts of the code

Analysis Results

Anomaly Name	**Analysis Findings**
Anomaly A	• **494** Vessels Analyzed • **16** Suspects (DFSI > 0)
Anomaly B	*0*, No anomalies after harbour activity checks
Anomaly C	**0**, No anomalies were detected
Anomaly D	• **494** Vessels Analyzed • **96** Identity Clone Events • **28** Suspects (DFSI > 0)

Challenges

Anomaly B:

- Requires pairwise comparisons ($O(N^2)$); this created a sequential bottleneck
- Used Latitude Banding and Early-Exits to skip unnecessary proximity checks between distant vessels
- The use of a global list for comparisons caused temporary RAM spikes
- Possible solutions include partitioning data into a $1x1$ degree grid to allow parallel processing per square

Anomaly vs. Noise (Harbor Activity):

- Added 12nm coastal geofencing and additional filters (e.g., min. drift checks ($> 300m$))

Profiling: Overhead and memory

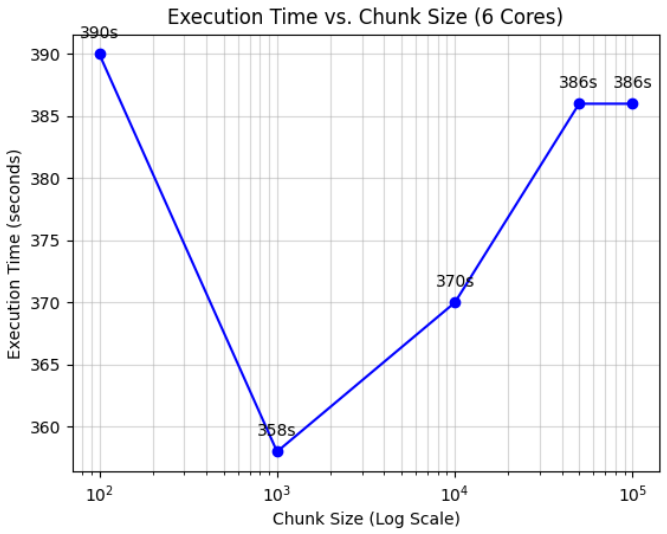


Figure: Chunk speedup

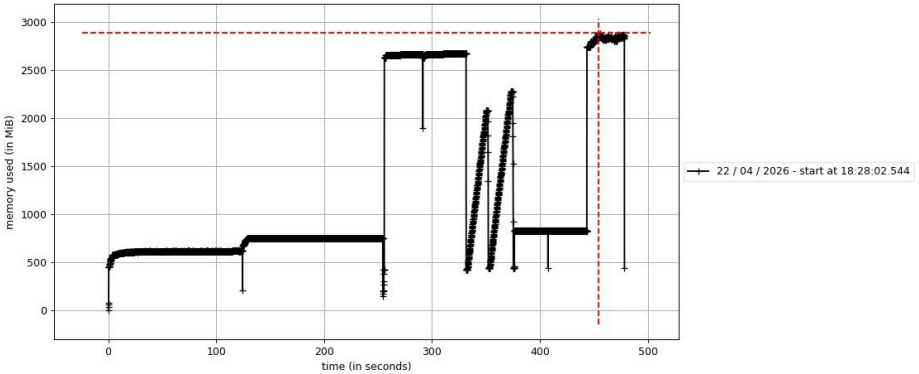


Figure: Total script memory Usage

Figure: Overall script overhead

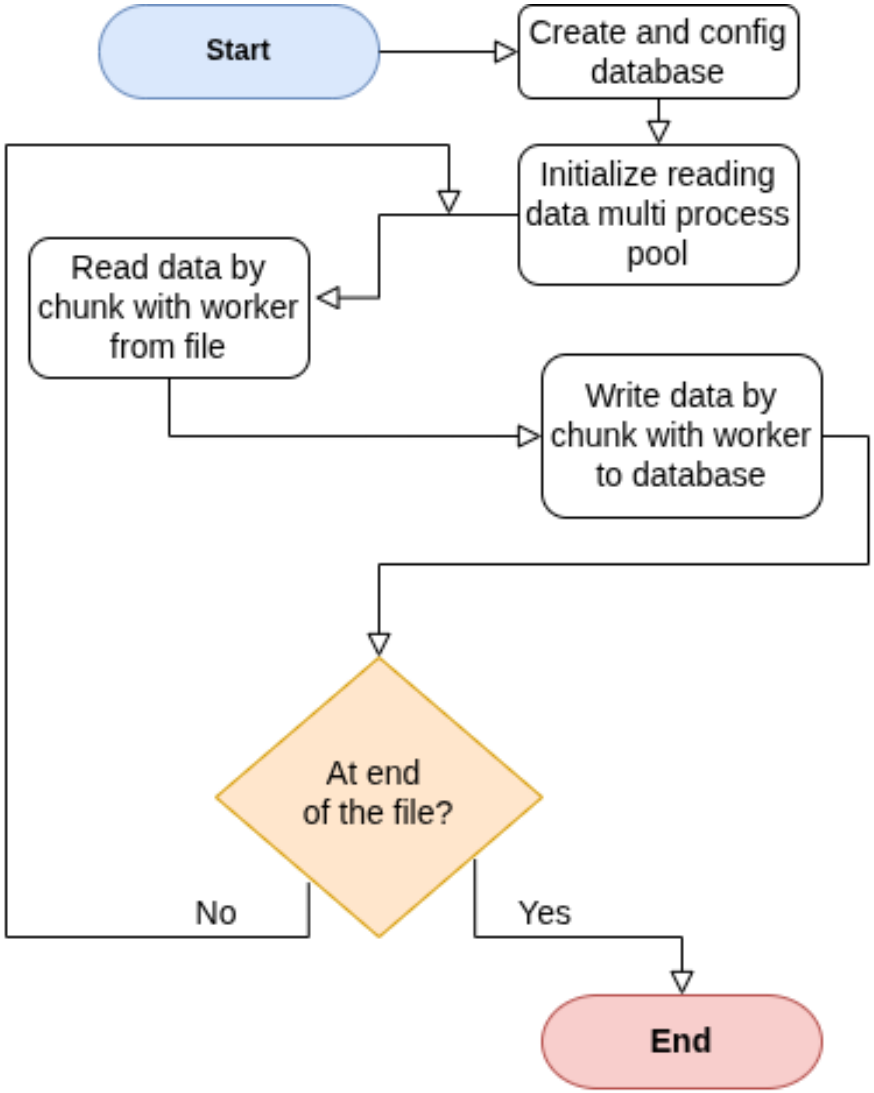
Visualizing Identity Cloning in the Baltic Sea

LAUNCH INTERACTIVE MAP

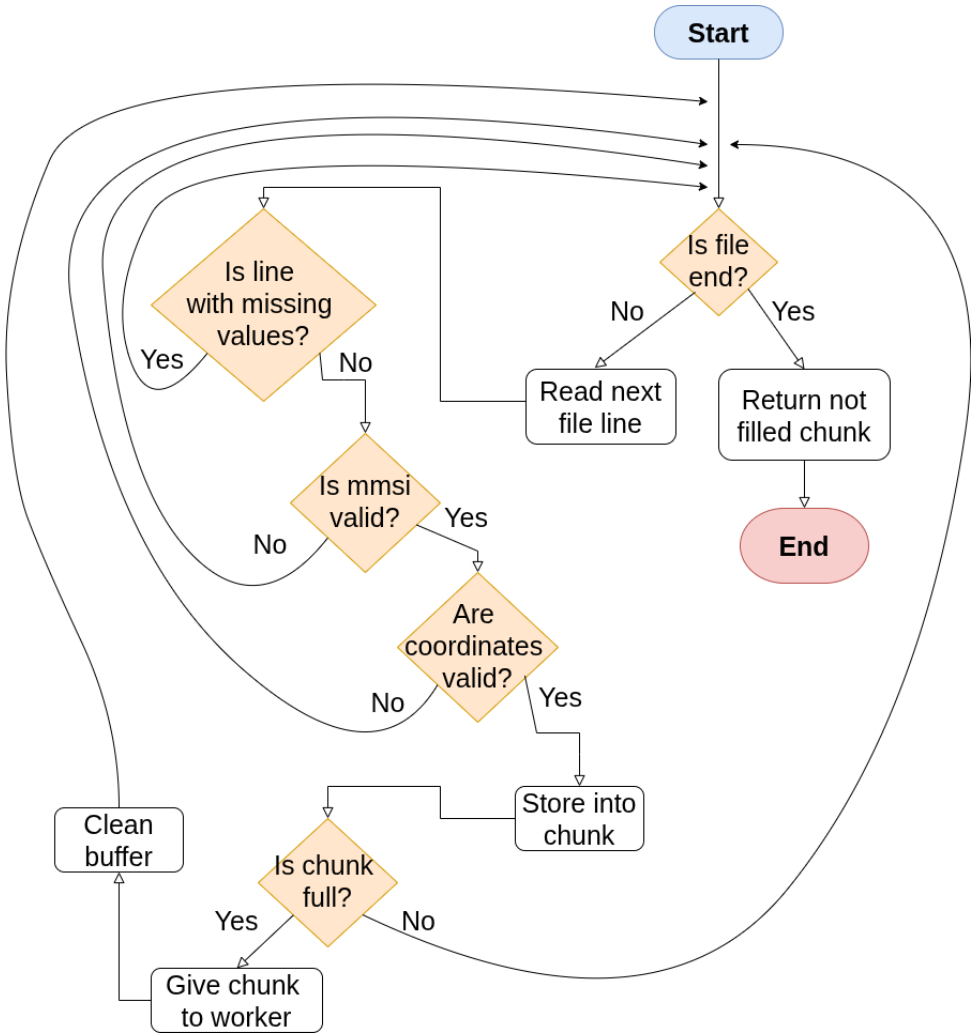
Map Geometry & Legend:

- • **Jump A:** The vessel's original reported GPS position.
- • **Jump B:** The "impossible" ping appearing shortly after.
- — **Jump Line:** The vector connecting two simultaneous locations.
- **Anomalous Speed:** Calculated velocity > 60 knots between points.

Appendix - Parser overview

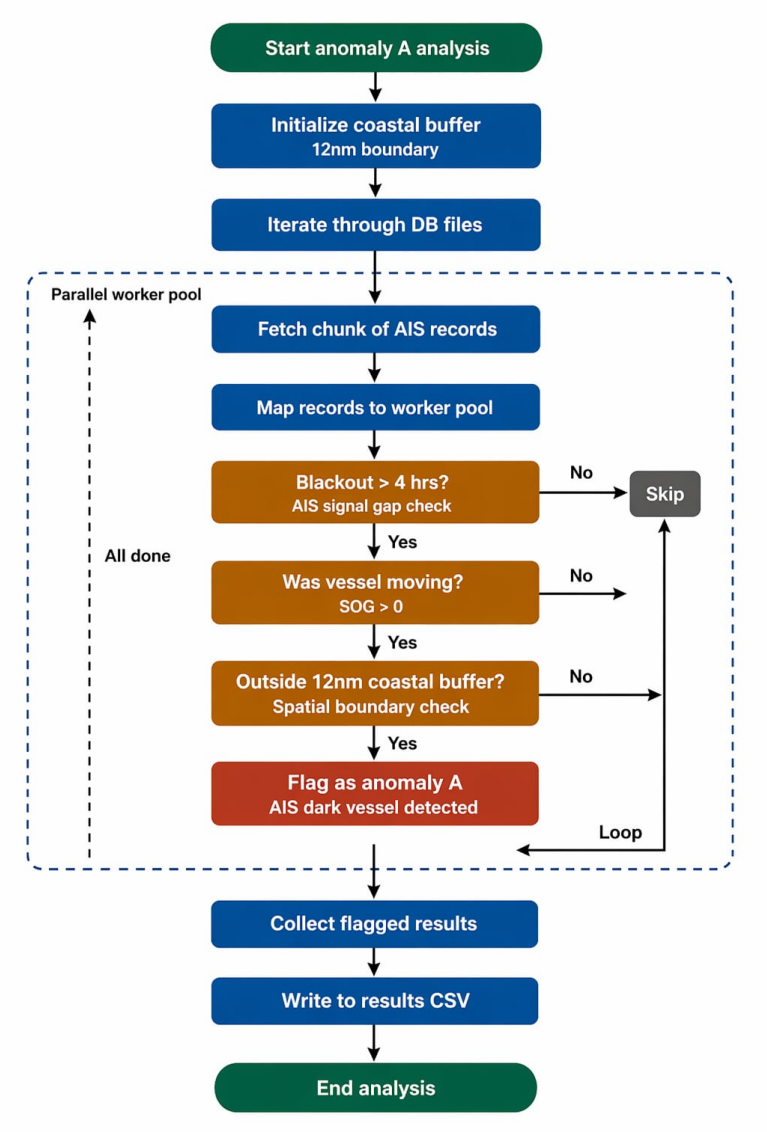


(a) Parser reader logic

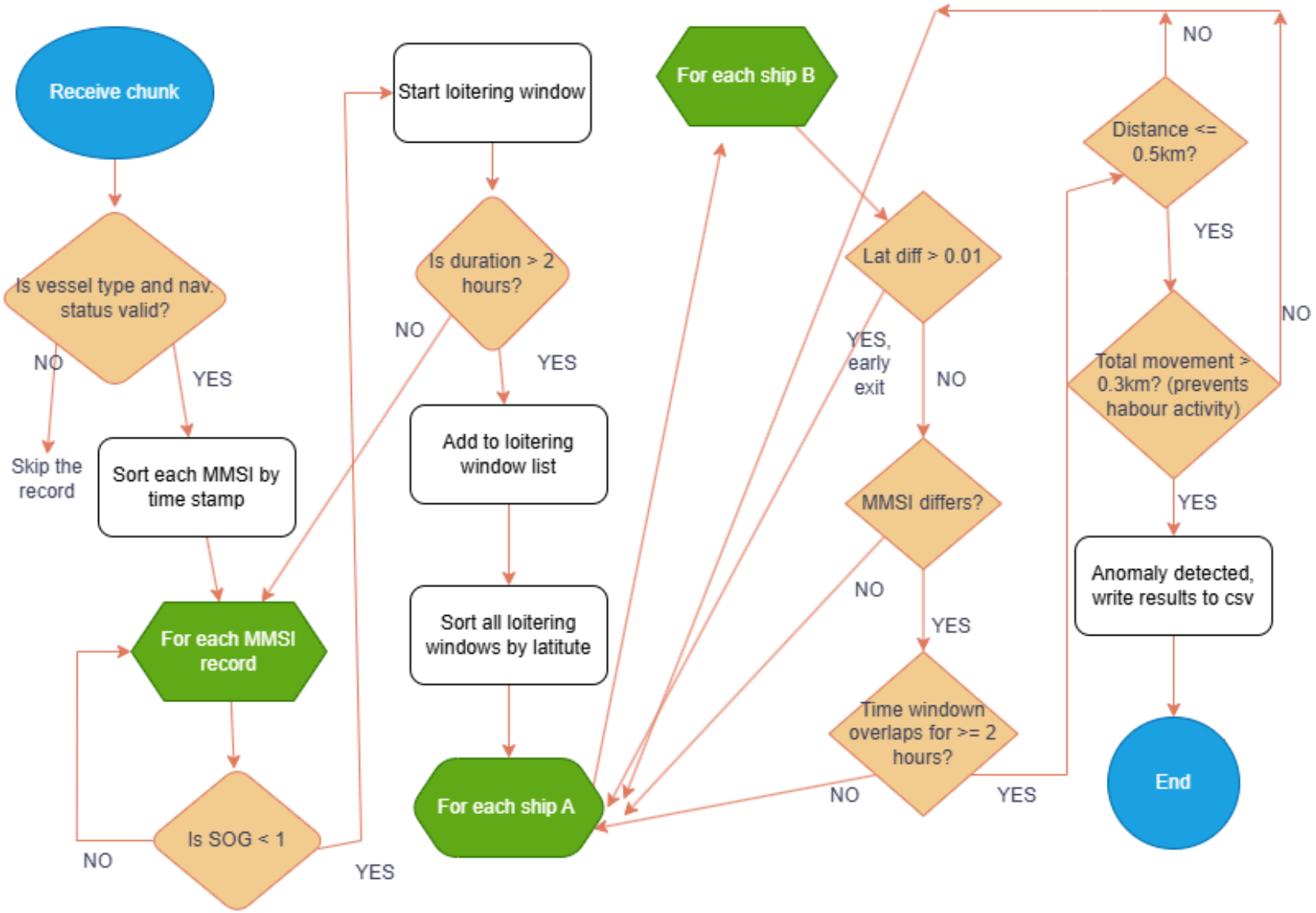


(b) Parser data validation

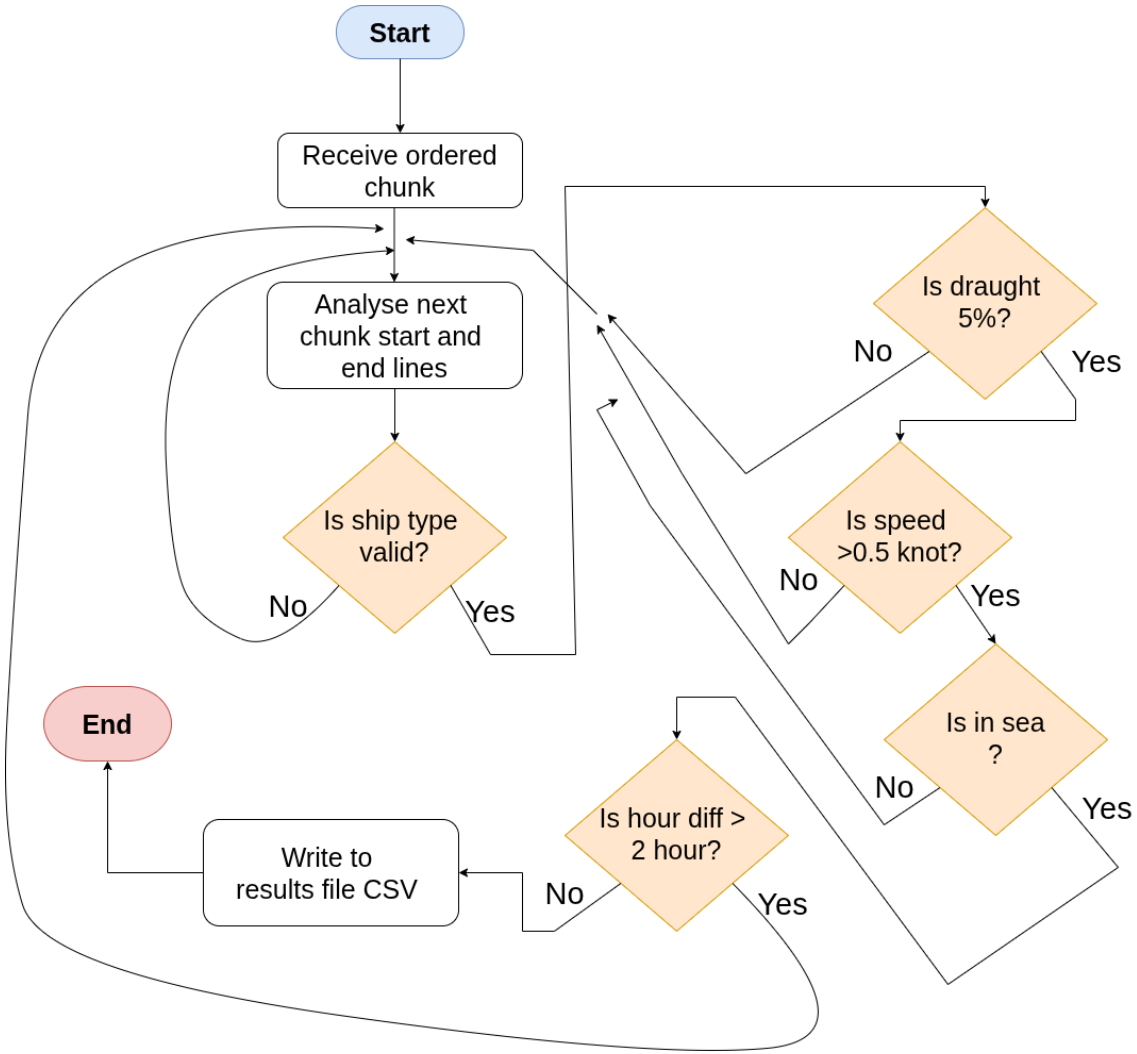
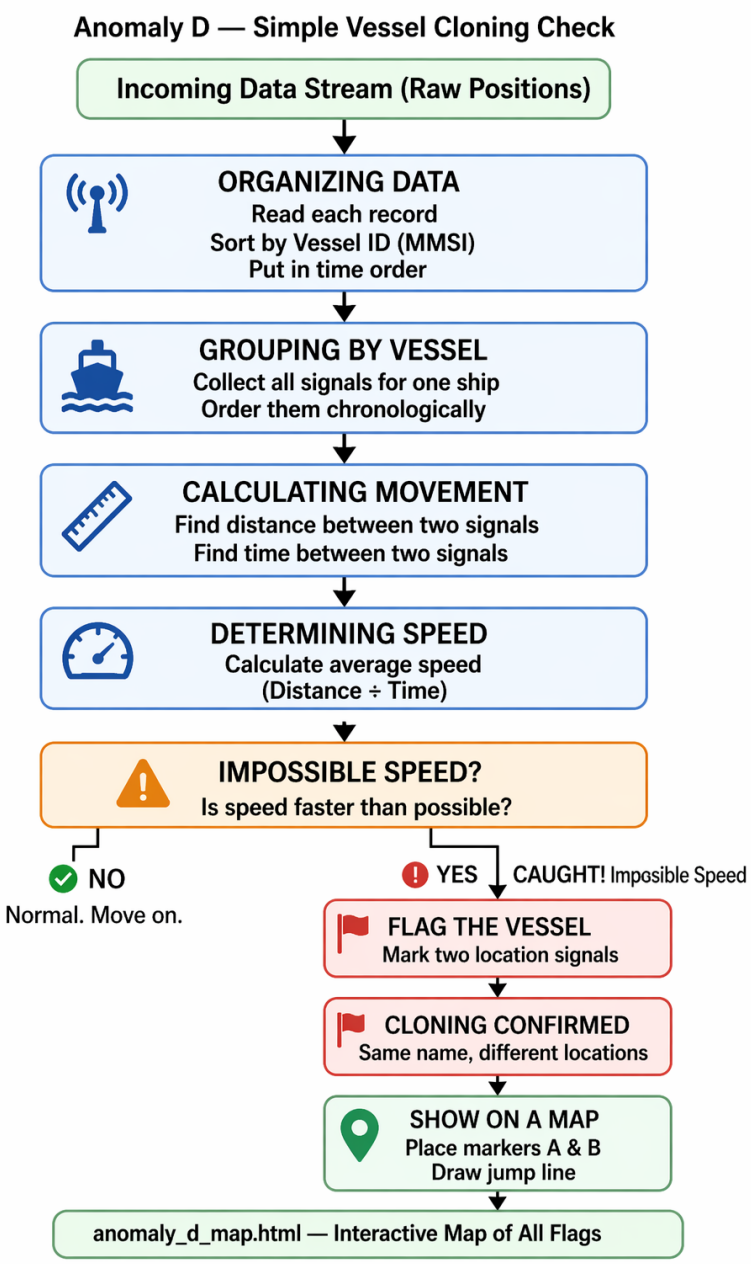
Appendix - Anomaly A detection



Appendix - Anomaly B detection



Appendix - Anomaly C detection



Appendix - Anomaly D detection